

ANALYSING THE IMPACT OF SOCIAL MEDIA ON SALES

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ABSTRACT:

The main objective of the project is to identify the relationship between social media engagement metrics and product or service sales. The study involves collecting real or sample datasets from various social media platforms and company sales records, followed by cleaning, processing, and analyzing the data using python.

❖ Python libraries such as Pandas and NumPy are used for data manipulation, Matplotlib and seaborn for data visualization, and Scikit-learn for implementing machine learning models like linear regression and decision trees to predict sales based on social media activity. Correlation analysis and time series analysis are also applied to uncover patterns and trends.

❖ By visualizing and interpreting these insights, the project aims to help businesses understand which social media strategies are most effective in driving sales. The outcome of this analysis can support data-driven decision-making, improve marketing ROI, and guide future digital marketing campaigns.

❖ This project not only demonstrates the practical application of data analytics in a real-world scenario but also highlights the growing importance of social media in shaping modern business strategies

INTRODUCTION:

The Role of social media in Business Growth:

- ❖ Social media has become a significant driver of brand visibility, customer engagement, and, most importantly, sales. With billions of active users on platforms like Facebook, Instagram, Twitter, and LinkedIn, businesses can reach a global audience and target specific demographics with tailored content. However, the real challenge lies in quantifying the effectiveness of social media efforts in terms of sales growth.
- ❖ Businesses use a variety of metrics to track social media performance, such as the number of likes, shares, comments, and overall engagement rates. While these metrics are useful for assessing the popularity of posts or the level of engagement, they do not directly translate to sales. Therefore, there is a need for robust data analysis methods that can help correlate social media metrics with actual sales figures, enabling businesses to optimize their marketing strategies.

SYSTEM REQUIREMENTS:

Hardware Requirements:

❖Processor:

A dual-core processor for handling basic tasks and At least 4 GB of RAM is required to work with data computation. However, for more complex data analysis efficiently. For larger datasets, 8 GB or more of RAM would tasks, a higher-end processor would improve performance.

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❖Storage:

Minimum of 2 GB of free storage space to store the project of files and data. Depending on the dataset size, additional space may be needed.

❖Graphics:

While not essential, a GPU can speed up the rendering of data visualizations, particularly for larger datasets.

Software Requirements:

❖Operating System:

The project can run on Windows, MacOS, or Linux, depending on the user's preference.

❖Python Version:

Python 3.8 or higher is recommended, as it includes enhanced features and libraries for data analytics.

❖Libraries:

▪Pandas:

Essential for data manipulation and cleaning. It used to load, analyze, and transform data into a usable format.

▪NumPy:

A library used for numerical operations, particularly working with arrays and performing mathematical operations.

▪Matplotlib & Seaborn:

For data visualization, these libraries help to create various charts to represent the findings.

▪Scikit-Learn:

Used for applying machine learning models to predict future sales trends and patterns based on historical data.

ARCHITECTURE:

- The architecture of analyzing the impact of social media on sales using python, Matplotlib, an dP and as involves several key steps that form a structured and insightful data analytics workflow.
- The process typically includes data import, data exploration, cleaning, grouping, and data visualization to uncover meaningful trends and relationships.

Let' s explore the architecture in more detail:

1. IMPORTING DA TA SET
2. DATA EXPLORATIO N
3. DATA CLEAN IN G
4. DATA GROUPIN G
5. DATA VISUALIZATION

IMPORTING DATASET:

Importing the data set is the first and most essential step in any data analysis project. In this stage, we bring data into the python environment from external sources like CSV files , Excel files, A PIs, or data bases.

- In this project, the dataset typically contains:

Social media engagement metrices.

Corresponding sales data for various products or campaigns.

- Python's panda's library is commonly used to import and manage the dataset.

Purpose:

To load the data in a structured form at (Data Frame), which allows for easy manipulation and analysis.

DATA EXPLORATION:

Data exploration, also known as exploratory Data Analysis (EDA), involves examining the data set to understand its structure, detect patterns, and identify potential issues.

■ Activities in this phase include:

Viewing the first few rows with `df.head()`.

Understanding column types and missing values using `df.info()`.

Getting statistical summaries with `df.describe()`.

Identifying unique platforms or engagement ranges using `df['platform'].unique()`.

■ Goals Of Data Exploration:

Gain familiarity with data set

Detect anomalies, such as extreme values or inconsistent formatting.

DATA CLEANING:

Data cleaning is the process of correcting or removing inaccurate records from the data set. This step ensures that the data is accurate, complete, and suitable for analysis.

• Common cleaning tasks:

Handling missing values: Using `dropna()` or filling with averages.

Removing duplicates: Using `drop_duplicates()`

Converting datatypes: For example, Converting a 'data' column to data time using `pd.to_datetime()`

Importance:

Clean data avoids misleading insights, improves the accuracy of visualizations, and enables more meaningful groupings and trends.

DATA GROUPING:

After the data is cleaned, grouping is used to organize and summarize data for analysis. Grouping helps break down large datasets into meaningful segments.

- In this project, data can be grouped:

- By platform (e.g., Facebook, Instagram) to compare their performance.

- By date to analyze trends over time.

- By campaign or content type to evaluate effectiveness

It helps identify which platforms generate the most engagement or sales, and allows tracking of performance overtime.

DATA VISUALIZATION:

Data visualization turns data into visuals, making it easier to understand insights, trends, and relationships. It is often the most impactful part of a data analysis project.

- Python libraries used:

- Matplotlib – for basic plots

- Seaborn – for advanced and statistical visualizations

- Common Visualizations:

- Bar Charts: Compare sales across platforms

- Line Charts: Show sales over time

- Pie Charts: Display distribution of engagement

- Heatmaps: Show correlation between variables

USES OF DATA ANALYSIS LIBRARY:

PANDAS

Data cleaning:

Handling missing or null values, removing duplicates, and transforming data into a more usable format.

Data manipulation: Sorting, filtering, and grouping data based on specific conditions or features.

Time Series Analysis: Pandas has strong support for handling time series data, which is crucial for analyzing sales trends over time.

NUMPY

Numerical computations:

It can handle complex mathematical calculations, including matrix operations and element-wise operations on arrays.

Statistical operations:

Calculating mean, median, variance, and standard deviation of data.

MATPLOTLIB & SEABORN

Matplotlib:

Provides basic plotting capabilities for visualizing relationships between variables.

Seaborn:

Built on top of Matplotlib, Seaborn allows for more advanced visualizations like heatmaps, distribution plots, and pair plots.

SCIKIT-LEARN

Data preprocessing:

Scaling, normalizing, and encoding data to make it suitable for machine learning models.

Modeling:

Implementing algorithms like linear Regression, Decision Trees, and Random Forests to predict sales based on social media metrics.

ADVANTAGES:

Insightful Decision Making:

By analyzing the impact of social media on sales, businesses gain valuable insights into which strategies or platforms are driving the most revenue. This allows them to make data-driven decisions rather than relying on gut feelings or assumptions.

Cost-effective Marketing:

By understanding which social media efforts are directly contributing to sales, businesses can optimize their marketing budget, focusing resources on high-impact platforms and campaigns, thus improving ROI.

Predictive Analytics:

Machine learning models can be used to predict future sales based on historical social media engagement. This predictive capability helps businesses plan ahead and adjust their strategies proactively.

Improved Marketing Strategies:

With concrete data linking social media performance to sales outcomes, businesses can refine their marketing strategies, tailor content to their audience, and adjust their posting schedules to maximize engagement and sales.

CONCLUSION:

- Data analysis using Python provides powerful tools to understand and quantify the impact of social media on sales. By analyzing various social media metrics and their correlation with sales data, businesses can optimize their social media strategies, improve customer engagement, and ultimately drive revenue growth. This approach allows for data-driven decisionmaking, leading to more effective marketing campaigns and a stronger return on investment.
- Data analytics can reveal the specific impact of different social media platforms and content types on sales.
- Techniques like A/B testing and correlation analysis can isolate the effect of social media campaigns from other marketing efforts.
- By measuring metrics like click-through rates, engagement, and conversion rates, businesses can quantify how social media contributes to revenue.
- Analyzing social media data helps identify which platforms and content formats are most effective for reaching the target audience and driving sales.
- This insight allows for the optimization of content calendars, ad spending, and targeting parameters.
- Social media analytics provides valuable insights into customer behavior, preferences, and sentiment.
- Sentiment analysis can reveal how customers perceive the brand and its products or services.

➤ This information can be used to tailor marketing messages, personalize customer experiences, and improve overall customer satisfaction.

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